MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 05 (SOLUTIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. In an AP, if $a = 1$, $a_n = 20$ and $S_n = 399$, then n is: [1]

(A) 19 (B) 21 (C) 38 (D) 42

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 38** [0.5 Mark]

Given: First term $a = 1$, last term $a_n = 20$, and sum of n terms $S_n = 399$.

We use the first-and-last-term sum formula of an AP:

Formula / Theorem Used: $S_n = \frac{n}{2}(a + a_n)$

Justification: [0.5 Mark]

$$399 = \frac{n}{2}(1 + 20) \implies 399 = \frac{21n}{2} \implies n = \frac{399 \times 2}{21} = 19 \times 2 = \mathbf{38}$$

Q2. The sum of the first 5 multiples of 3 is: [1]

(A) 45 (B) 55 (C) 65 (D) 75

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 45** [0.5 Mark]

Given: The first 5 positive multiples of 3 are $3, 6, 9, 12, 15$, which form an AP with $a = 3$, $l = 15$, and $n = 5$.

We apply the sum formula:

Formula / Theorem Used: $S_n = \frac{n}{2}(a + l)$

Justification: [0.5 Mark]

$$S_5 = \frac{5}{2}(3 + 15) = \frac{5}{2} \times 18 = 5 \times 9 = \mathbf{45}$$

Q3. If the sum of n terms of an AP is $S_n = 3n^2 + 5n$, then its common difference d is: [1]

(A) 3 (B) 6 (C) 5 (D) 2

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(B) 6** [0.5 Mark]

Given: $S_n = 3n^2 + 5n$. We recover a_1 and a_2 from partial sums:

Formula / Theorem Used: $a_1 = S_1, \quad a_2 = S_2 - S_1, \quad d = a_2 - a_1$

Justification: [0.5 Mark]

$a_1 = S_1 = 3(1)^2 + 5(1) = 8$, and $S_2 = 3(2)^2 + 5(2) = 12 + 10 = 22$.

Hence $a_2 = S_2 - S_1 = 22 - 8 = 14$, giving $d = a_2 - a_1 = 14 - 8 = \mathbf{6}$.

Q4. For what value of k will $k, 2k-1, 2k+1$ be consecutive terms of an AP? [1]

(A) 2 (B) 3 (C) -3 (D) 1

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(B) 3** [0.5 Mark]

We apply the condition for three consecutive terms a, b, c to be in AP:

Formula / Theorem Used: $b - a = c - b \iff 2b = a + c$

Justification: [0.5 Mark]

$2(2k-1) = k + (2k+1) \implies 4k - 2 = 3k + 1 \implies 4k - 3k = 1 + 2 \implies k = \mathbf{3}$.

Q5. **Assertion (A):** The sum of the first 10 natural numbers is 55.

Reason (R): The sum of the first n natural numbers is $S_n = \dfrac{n(n+1)}{2}$. [1]

(A) Both (A) and (R) are true; (R) explains (A). (B) Both are true; (R) does not explain (A).

(B) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true; (R) explains (A).** [0.5 Mark]

Using the AP sum formula for $1, 2, 3, \ldots, n$ ($a = 1, d = 1$):

Formula / Theorem Used: $S_n = \dfrac{n}{2}[2(1) + (n-1)(1)] = \dfrac{n(n+1)}{2}$

Justification: [0.5 Mark]

Reason (R) gives the exact AP sum identity (true). Putting $n = 10$ gives $S_{10} = \frac{10 \times 11}{2} = 55$, so Assertion (A) is true and directly explained by Reason (R).

Q6. How many terms of the AP $24, 21, 18, \ldots$ must be taken so that their sum is 78? Explain the double answer. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Setting Up the Quadratic Equation [0.5 Mark]

Given: AP is $24, 21, 18, \ldots$, so $a = 24$, $d = 21 - 24 = -3$, and $S_n = 78$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Step 2: Solving for n by Middle-Term Splitting [1 Mark]

$$\frac{n}{2}[2(24) + (n-1)(-3)] = 78 \implies \frac{n}{2}[48 - 3n + 3] = 78 \implies n(51 - 3n) = 156$$

Dividing throughout by 3:

$$n(17 - n) = 52 \implies n^2 - 17n + 52 = 0$$

Splitting the middle term ($-4 - 13 = -17$, $(-4)(-13) = 52$):

$$n^2 - 4n - 13n + 52 = 0 \implies n(n-4) - 13(n-4) = 0 \implies (n-4)(n-13) = 0 \implies n = 4 \text{ or } 13$$

Step 3: Explaining the Double Answer [0.5 Mark]

Since $d = -3 < 0$, terms decrease and turn zero and then negative: $a_5 = 12, a_6 = 9, a_7 = 6, a_8 = 3, a_9 = 0, a_{10} = -3, a_{11} = -6, a_{12} = -9, a_{13} = -12$. Thus the sum of terms from a_5 to a_{13} is zero ($a_5 + \cdots + a_{13} = 0$), so both 4 terms and 13 terms sum to 78.

$$\boxed{\mathbf{n = 4 \text{ or } n = 13}}$$

Q7. If the nth terms of the two APs $9, 7, 5, \ldots$ and $24, 21, 18, \ldots$ are the same, find the value of n. Also find that term. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: First AP: $9, 7, 5, \ldots$ ($a = 9$, $d = 7 - 9 = -2$); Second AP: $24, 21, 18, \ldots$ ($A = 24$, $D = 21 - 24 = -3$).

We use the general term formula of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Writing the nth Term of Both APs [1 Mark]

For AP 1: $a_n = 9 + (n-1)(-2) = 9 - 2n + 2 = 11 - 2n$.

For AP 2: $A_n = 24 + (n-1)(-3) = 24 - 3n + 3 = 27 - 3n$.

Step 2: Equating $a_n = A_n$ and Finding n [1 Mark]

$$11 - 2n = 27 - 3n \implies 3n - 2n = 27 - 11 \implies n = 16$$

Step 3: Finding the Common 16th Term [1 Mark]

$a_{16} = 11 - 2(16) = 11 - 32 = -21$ (Check: $A_{16} = 27 - 3(16) = 27 - 48 = -21$).

$$\boxed{\mathbf{n = 16}, \quad \mathbf{a_{16} = -21}}$$

Q8. For what value of p are $2p-1$, 7, and $3p$ three consecutive terms of an AP? Also find the next term of the AP. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: The numbers $2p-1$, 7, and $3p$ are three consecutive terms of an AP.
Using the arithmetic mean property:

Formula / Theorem Used: If a, b, c are in AP, then $2b = a + c$

Step 1: Forming and Solving Equation for p [1.5 Marks]

$$2(7) = (2p-1) + 3p \implies 14 = 5p - 1 \implies 5p = 15 \implies p = \frac{15}{5} = 3$$

Step 2: Forming the AP and Finding the Next Term [1.5 Marks]
Substituting $p = 3$ into the three terms: $2(3) - 1 = 5$, 7, and $3(3) = 9$.
Here $a = 5$ and common difference $d = 7 - 5 = 2$. Next (4th) term $= 9 + 2 = 11$.

$$\boxed{\mathbf{p = 3}, \quad \text{Next term} = \mathbf{11}}$$

Q9. Find the sum of the first 15 multiples of 8. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Forming the AP and Identifying Parameters [1 Mark]
Given: First 15 multiples of 8 are $8, 16, 24, \ldots, 120$. This forms an AP with $a = 8$, $d = 16 - 8 = 8$, and $n = 15$.
We use the sum of n terms formula:

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Step 2: Substituting Values and Simplifying [2 Marks]

$$S_{15} = \frac{15}{2}[2(8) + (15-1)(8)] = \frac{15}{2}[16 + 14 \times 8] = \frac{15}{2}[16 + 112]$$

$$S_{15} = \frac{15}{2} \times 128 = 15 \times 64 = \boxed{\mathbf{960}}$$

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Q10. CASE STUDY: Step-Ladder Tapering Rungs (Carpentry)

A carpenter constructs a wooden step-ladder whose horizontal rungs are spaced uniformly 25 cm apart from centre to centre. The lengths of the rungs decrease uniformly from 45 cm at the bottom rung to 25 cm at the top rung. If the vertical distance between the top and bottom rungs is 2.5 m (250 cm): [4]

(i) Calculate the total number of rungs needed in the ladder. [1]

(ii) Determine the common difference d of the rung lengths from bottom to top. [1]

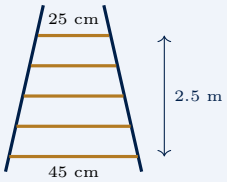
(iii) Compute the total length of wood required to fabricate all the rungs. [2]

— **OR** —

Find the exact length of the middle rung of the ladder. [2]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Extracted Scenario Data: Bottom rung $a_1 = 45$ cm, top rung $a_n = 25$ cm, span $= 2.5$ m $= 250$ cm, gap $= 25$ cm.
We apply the AP term and sum formulas:



Formula / Theorem Used: $n = \frac{\text{Span}}{\text{Gap}} + 1, \quad a_n = a + (n-1)d, \quad S_n = \frac{n}{2}(a + l)$

Sub-part (i): Total Number of Rungs [1 Mark]

Since there is one rung at each endpoint of the 250 cm span:

$$n = \frac{250}{25} + 1 = 10 + 1 = \boxed{\mathbf{11} \text{ rungs}}$$

Sub-part (ii): Common Difference (d) Bottom to Top [1 Mark]

With $a_1 = 45$ and $a_{11} = 25$:

$$45 + (11-1)d = 25 \implies 10d = 25 - 45 = -20 \implies d = \boxed{\mathbf{-2} \text{ cm}}$$

Sub-part (iii): Total Length of Wood Required for Rungs [2 Marks]

Using $S_{11} = \frac{n}{2}(a + l)$ with $n = 11$, $a = 45$ cm, and $l = 25$ cm:

$$S_{11} = \frac{11}{2}(45 + 25) = \frac{11}{2} \times 70 = 11 \times 35 = \boxed{\mathbf{385} \text{ cm} \quad (\text{or } 3.85 \text{ m})}$$

Sub-part (iii) [OR Option]: Length of the Middle Rung [2 Marks]

Since $n = 11$ is odd, the middle rung is the $\left(\frac{11+1}{2}\right)^{\text{th}} = 6^{\text{th}}$ rung.
Using $a = 45$ cm and $d = -2$ cm:

$$a_6 = 45 + (6-1)(-2) = 45 - 10 = \boxed{\mathbf{35} \text{ cm}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★